



AJYAD
— THERMOTECH —



AJYAD FOR REFRACTORIES

AJYAD for develops refractories

PROTECH develops and produces refractory products for the steel industry. We offer a full range of unshaped products.

PROTECH offers refractory solutions from the completely fireproof monolithic raw material mix up to the fully assembled vacuum snorkel with integrated argon gas pipes.

Our philosophy: We supply refractory concepts, tailor-made to the individual customer's demands. We recognise that every steel mill is unique. Each plant needs an individual solution – and this must be innovative and cost efficient.



Your steel is worth it.



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Applications of EBT Sand

Olivine sand, due to its high melting point and sinter, low heat loss and resistance to thermal shocks has applications in refractory and steel manufacturing industries. Some of such applications are:

- Usage for filling the EBT value of the electric arc furnaces
- Application in producing Tandish cover.
- Application in producing magnesium-chromium bricks and low-magnesium bricks
- For covering the back wall in the AOD convertors.



EBT OLIVINE SAND	
Chemical Analysis	
MgO	47-50 %
CaO	0.5-1 %
SiO ₂	39%
Fe ₂ O ₃	6-10 %
Type of Bonding,	Chemical
Grain Size, mm	2-6mm

DRY BACKFILL MIXES

PROTECH has developed the FOR RAM series dry lining mixes for avoiding infiltrations and running behind the bricklaid safety lining. These are high quality alumina- or magnesite-based ceramically bonding products.

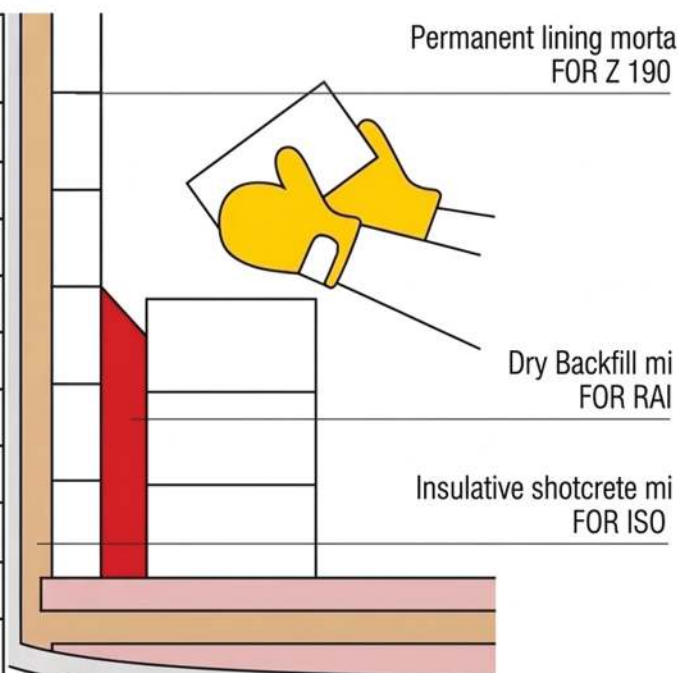
The dry lining mixes are easy to process without any further aid. They are simply poured between the brick rings and the safety lining during the bricklaying of the wall and rammed by hand.

DRY BACKFILL MIXES

Chemical Analysis

MgO	90%
CaO	3-5%
SiO ₂	5%
Fe ₂ O ₃	1-3%
Type of Bonding,	Chemical
Grain Size, mm	0-3mm
<u>Physical Properties</u>	
Refractoriness °C	1610
Max Service Temperature, °C	1560

Physical Properties



Silica Ramming Mass

PROTECH has developed the Silica Ramming Mass is used in the lining of induction furnaces. Quality of Acidic Ramming mass is directly related to the heating performance 17-18 heats of the furnaces. Better quality of lining results in the smooth working of furnaces, optimum output and better metallurgical control. It is manufactured under strict supervision of qualified & experienced work force. Proper check is exercised on magnetic process to assure from free material and to assure proper composition of granules according to furnace capacity. The granulation of the mix is optimized keeping in view the furnace makes and capacity, thus giving maximum and best results. It is characterized by thermal stability, corrosion resistance and wear resistance because it contains less binders, fire clay and moisture compared to plastic refractories.



Silica Ramming Mass	
Application,	Dry Ramming
Chemical Composition.	
Moisture	≥ 0.10 %
SiO ₂	98.00%
Fe ₂ O ₃	≥ 0.01 %
H ₃ BO ₃	≥ 1.30 %
Physical Properties.	
Method of Application,	Ramming
Max Service Temperature, °C	1715
GRAIN SIZE	any size

BLAST FURNACE

In the blast furnace there are more monolithic refractory products in continuous operation than at any other stage in the steel production process - excluding new linings and interim repairs. Taphole mixes, runner mixes, assembly parts or preshapes (for example skimmer, tipping trough) are heavily-used refractory products in the production of pig steel.

Pig iron is usually transported to the steel mill in special containers (torpedo ladle, pig iron ladle), which can vary considerably from plant to plant. Therefore, individual, creative refractory solutions are required - from highly adhesive mortars to insulation and maintenance mixes.

PROTECH supplies the whole range of monolithic refractory products for the blast furnace including: shaft maintenance, mux masses, light heat and erosion proof gunning mixes or tapping hole masses, offered through cooperation with world-leading technology partners.

PROTECH monolithic refractories for the blast furnace cover:

- gunning mixes
- tap hole mixes
- castables
- mortar
- preshapes



Hearth Ramming Mass

Hearth Ramming Mass is a magnesia-based materials, that designed and produced to be applied by cold ramming forming strong ceramic bond through heating to serve at bottom of EAF facing impacting, wear, and abrasion at elevated temperatures 1750 °C.



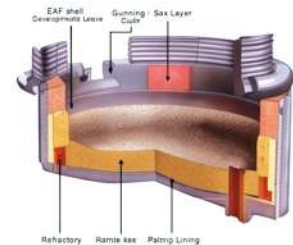
ProRam-G	
Grain Size, mm	0 – 8
Chemical Properties.	
MgO	80 – 85 %
SiO₂	1 – 2 %
Fe₂O₃	1.5 %
Al₂O₃	0.8 %
CaO	15 %
Physical Properties.	
Method of Application,	Cold Ramming
Max Service Temperature, OC	1750
Grain Density (g/cm³)	3.20
Type of Bonding	Ceramic
Expiration, Months	12

Hot Fettling Mass

Hot Fettling Mass is a magnesia-based monolithic that designed and produced to be applied by hot spraying using fettling machine to cover the arch-sided and as repairing for Hearth Ramming Mass at the bottom of EAF, it is chemically adjusted to form that ceramic bond during heating up, its GSD is designed to be easily sprayed and applied.

PT1700-85L	
Grain Size, mm	0 – 8
Chemical Properties.	
MgO	80 – 85 %
SiO₂	1 – 2 %
Fe₂O₃	1.5 %
Al₂O₃	0.8 %
CaO	5 – 10 %
Physical Properties.	
Method of Application,	Hot Repair
Max Service Temperature, °C	1750
Grain Density (g/cm³)	3.20
Type of Bonding	Ceramic
Expiration, Months	12

Hot Gunning Mass



Magnesia Hot Gunning Mass is a material produced and mixed with special binders that give it stickability properties to be used as a hot repairing material of the EAF wall for any leakage and/or cracks.

MgPT-90H	
Grain Size, mm	0 – 3
Chemical Properties.	
MgO	88 – 92 %
SiO ₂	3.5 %
Fe ₂ O ₃	1.5 %
Al ₂ O ₃	0.8 %
CaO	2.5 %
Physical Properties.	
Method of Application,	Hot Repair
Max Service Temperature, °C	1750
Bulk Density (g/cm ³)	2.50
Type of Bonding	Ceramic
Expiration, Months	12

Pre-Shaped Castables

Pre-Shaped Castables that special shapes of refractories designed and fabricated to be served in flow system of molten steel.

It is designed chemically at first to resist highly abrasion of flow molten steel and slag reaction, then casted using highly technique and dried very carefully to give the maximum properties.

Our company has wide and very good experience in production and fabrication of such products.



Tundish Spray Mass



Tundish Spray Mass is a Magnesia-based material designed and graded less than 1 mm to be sprayed as working layer for tundish that serve many purposes during and after casting.

PT5001-HM	
Chemical Properties.	
MgO	90.0 %
CaO	3.0 %
Fe ₂ O ₃	1.5%
SiO ₂	5.0 %
Physical Properties.	
Grain size	0-0.5 mm
Bulk Density	1.4 – 1.7 g/cm ³
Expiration	12 Months



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